Its GDP has grown from 1.7 percent in 2012 to 3.3 percent in 2014, and is expected to grow by 3.2 percen in 2015 and 3.3 percent in 2016.

The EC and the prospective project co-financiers have confirmed their preference for the Bank leadership of the new large investment program, with the partial aim to use the program to bring the institutional innovations into practice.

Many of these centers have been located along transport corridors and, effectively, in former floodplains and in Poland's most productive agricultural area, with high added-value from orchards and horticulture.

and economically productive areas and assets such as factories, commercial centers, touristic locations, strategic transport and communication networks, farms, etc Thus, the project indirectly supports shared prosperity and job creation benefiting all of the identified beneficiary population.

The proposed project will further strengthen the national institutional capacity for flood management and forecasting, as well as improve the capabilities to operate existing and new infrastructure.

The project will also help to strengthen the capacity to prepare RBMPs and investment prioritization plans, improve the flood policy framework, and enhance the capability for more reliable forecasting, early warning, and infrastructure operation of existing and new infrastructure.

The activities will include the reconstruction of dikes and other bank protective works (revetments, parapets, and so on), dredging in the Odra River as well as in canals and the harbor of Szczecin, and river training works, that is, the recalibration and (re)construction of groynes and lateral submerged dams in the river, restoration of bends, and protection of banks

The works comprise of (i) the reconstruction and extension of dikes and embankments along the Vistula to replace old unreliable dikes; (ii) bank stabilization and strengthening with rip-rap, revetments, etc.;

Through Component 4 additional support will be provided for the preparation of the main parts of the RBMP and the investment prioritization plan for the Upper Vistula.

The studies will, among other things, cover the preparation of follow-up investments and the development of a project-based communication strategy.

The Odra River Basin Flood Protection Project is co-financed by the CEB and the EC (Cohesion Funds) which provide parallel funding within the framework of the program and investment design parameters set by the Bank.

This compares favorably with other large hydraulic investments such as the construction of the Swinna-Poremba reservoir that after 25 years is not completed yet.

Finally, it is universally accepted that there is a need to move from an exclusive infrastructure approach to a comprehensive and holistic approach to flood management combining infrastructura investments with management and policy measures

A dedicated survey platform will be developed to allow impact evaluation of floods and flood mitigation efforts, to better analyze the disaggregated impact of the project as well as investment programs (disaggregation by income, location, social variables, gender, and so on) and enable better targeting of future investments and measures. A

Secondly, the experience with the ongoing Odra River Basin Flood Protection Project has shown that the government is committed to provide the budgets for investment as well as for operation and maintenance (O&M).

However, with the expansion of the infrastructure, the need to increase the O&M budget is growing apace.

This risk is mitigated, partly, by the following steps: (i) the extensive screening of investment proposals that has taken place over the past years by the prospective Implementing Agencies; (ii) the requirement that, for local situations where flood defense could be achieved through different alternatives (for example, through a flood by-pass, higher dikes, local overflow areas, or combinations) the alternatives have been exhaustively compared; and (iii) by the availability, since early 2014 of the draft Flood Risk Maps, that form the basis for the Updated Master Plans that have been completed in 2014 and re-confirm the prioritization of the project selection.

Generally, costs can only be recovered indirectly through increases in land and residential taxes, and the higher general public revenue generated by the improved investment climate and business activity. The private sector generally contributes to flood protection only through the insurance services to compensate households and businesses that could be affected. In Poland, the private sector participates intensively in flood projects through the assignments for design, consulting, contracting, and maintenance of the infrastructure.

The economic rate of return (ERR) of the project is 13.5 percer The Net Present Value (NPV) of the overall project is estimated at PLN 4,815 million (with 5 percent as discount rate) and at PLN 926 million (with 10 percent as discount rate)7 .

The estimated ERR is robust and not very sensitive to costs and benefits: even a 30% increase in investment costs, or up to 40% percent reduction of benefits from avoided material flood damage would maintain an ERR above 10 percent.

The main benefits quantified for this assessment are related to (i) the reduced material an non-material damages from flood hazards; (ii) induced economic activities including mining aggregate; (iii) enhanced navigability of the Odra River; (iv) increased tourism due to improve safety conditions in the area; and (v) jobs created during the implementation of investments.

In the Middle and Lower Odra, several natural areas alongside the river still give the river space to expand; immediately upstream of Szczecin the vast Miedzyodrze wetland, situated between the two parallel Odra channels, fulfills the same function.

The project would fund investments in medium-scale overflow polders that are located in strategic locations, but will not yet provide the final required space for flood wave absorption.

These studies take also into account the expected influence of climate change and land-use change, however, future investments, especially of more strategic structures, would require a deepening of such analysis and simulation.

In many cases the works involve the rehabilitation of existing infrastructure in mostly government-owned lands located in sparsely populated peri-urban and rural areas (e.g.

Based on the early conceptual designs for subprojects to be built in 2019 (Component 2), there is also the possibility that some additional 30 households (approximately 100 people) may also require resettlement.

At the broader level, the preparation of the River Basin Management Plans (RBMPs) and prioritization of investments will be based on multi-stakeholder consultations (e.g.

It is important to note that the majority of the investments concern rehabilitation and modernization of already existing structures.

The project will also finance the construction of several overflow areas (dry polders).

Project Developmen

PDO Statement

The project developm
Vistula River basins a

These results are at

Project Developmen

Indicator Name	Core
Total area in the 1% flood plain benefiting from enhanced protection and operational forecasts	

[8] Based on current 1% floo
[9] Estimated for horizon to `

Total population benefiting from enhanced protection and operational forecasts (gender disaggregated)	
Flood Operation Centers established and functional	

Intermediate Result

Indicator Name	C
Component 1A: Length of enhanced protection in the	

[10] Based on current flood n
[11] Estimated for horizon to

Zachodniopomorskie Province
Component 1B: Modernizing and reconstructing the middle and lower Odra river systems to upgrade to a Class III waterway
Component 1C: Extension and construction of flood embankments to protect Słubice City
Component 2A: Construction of dry polders in the Nysa Kłodzka Valley
Component 2B: Length of enhanced protection in the Nysa Kłodzka Valley

[12] This component represer

Component 3A: Construction of dry polders to protect Upper Vistula towns and Kraków	
Component 3B: Length of enhanced protection for Sandomierz and Tarnobrzeg	
Component 3C: Length of enhanced protection for the Raba sub-basin	
Component 3D: Length of enhanced protection for the San sub-basin	
Component 4B: Flash flood systems for sub-basins operational	

RBMPs or Investment Prioritization Plans drafted for Vistula sub-basins	
% of project-supported RBMPs and investments informed by citizen feedback through consultations (disaggregated by gender)	

In Western Poland, the Odra benefited from ample government attention starting in the 19th century and this had led to major investments in river navigability and flood protection in the early 20th century.

After 1945, investments in flood protection took a back seat as the country was rebuilding its cities and its industrial base

It is oriented toward structural and nonstructural flood protection measures which are designed to gradually increase the level of flood safety of people as well as economic and cultural assets in the Upper Vistula basin with a time horizon of 2030.

It also received access to large EU funding allocations (Structural and Cohesion Funds) that can finance up to 85 percent of eligible infrastructure and other measures

Flood Management Investments and Measures Prioritization

16.

The project will also be

[1] Flood damage relates to the physical and non-physical costs for reconstruction and repairs of infrastructure and other assets to restore livelihoods to the situation before the flood.

Investments in

upstream retention, generally, can also lead to lower investments downstrear

For several proposed works and measures, the economic rationale is confirmed and advanced documentation for works exists, yet for other sets of investments additional studies to optimize the investments will be required before further decisions can be made on their meri The project will finance selected investments as well as studies that have the purpose to analyze different investment scenarios

The Warta was not prioritized because it has a low readiness to proceed, which is the consequence of the presence of the large Natura 2000 protected forest tracts that, for the momen preclude investments in flood management.

On the other hand, the preparation of such a long-list of meritorious investments and measures had to be initiated for the Upper Vistula. While some of these studies will be completed in 2015 or 2016, they now provide a robust basis to discern investments that can be considered no-regret, and (strings of) investments where more extensive analysis is required to decide on the variant

In particular, these analyses investigate the net effect of new infrastructure on the string of downstream and upstream locations.

In these instances, the project will support financing for in-depth studies but not the investments in works.

Firstly, many of the old dike and embankment systems near or around towns and infrastructure need to be modernized and/or expande

This, in turn, will help to reach the WFD goals, and it will also significantly enhance the tourist potential of the region, generating jobs.

The works will comprise construction of 2.3 km of flood embankments with overpasses, anti-filtration membrane, provision of mobile flood screen for one location, gabion protection, and ancillary works to protect two towns from Odra water backflow.

The works will include the reconstruction of the 3.49 km long dike with protective screens and a top road, as well as the modernization of the Krajnik pumping station.

The works will include the dredging and stabilization of 21 channels; reconstruction of 32 automatic gates, locks, and passes; levelling of 60 km of dike tops; and provision of 32 footbridges.

The task includes the reconstruction of regulating structures and groynes, lateral breakwaters, revetments along banks, bank stabilization, minor dredging, and so on.

.B.3 Construction of docking/mooring infrastructure on the Lower Odra and Border Odra including the new marking of the shipping lane.

This

subcomponent includes the following proposed works:

1.C.1 Extension and construction of flood embankments.

These may be

included in the first 18 month program (following confirmation) and procurement initiated for these investments (four contracts) with the aim to mobilize the contractor during 2016.

As part of the entire work, in particular, the following will be performed: section-based modification and renovation of the existing bank protection measures; construction of new embankments and floodwalls on the section whose total length is 14.5 km; modification of the existing embankments and floodwalls on the section for a total length is 6.5 km; enhancement of throughput of 38 bridge and footpath structures; and enhancement of throughput of 13 small weirs and barrages.

As part of the entire work, in particular, the following will be performed: section-based modification and renovation of the existing bank protective measures and enhancing the throughput of the two river beds; construction of new embankments and floodwalls on the section with a total length of 25 km; modification of the existing embankments and floodwalls on the section whose total length is 4 km; enhancement of throughput of 23 bridge and footpath structures; and enhancement of throughput of 9 minor weirs and barrages.

As part of the entire work, in particular, the following will be performed: section-based modification and renovation of the existing bank protective measures and enhancing the throughput of the beds of the two rivers; construction of new embankments and floodwalls on the section with a total length of 8 km; modification of the existing embankments and floodwalls on the section with a total length of 6.5 km; enhancement of throughput of 66 bridge and footpath structures; and enhancement of throughput of 12 minor weirs and barrages

The Raba tributary, and, inside that sub-basin, selected coherent investments that were prioritized based on a technical, social, and economic ranking of strings of investments

4. The San tributary, and, inside that sub-basin, selected coherent investments that were prioritized based on a technical, social, and economic ranking of strings of investments

46. Though the Raba and San rank high in priority among all right-bank sub-basins and command priority attention, they may not necessarily yield the highest economic returns when all benefits are calculated across the whole basin and for all possible precipitation and investment scenarios.

Expansion of embankments protecting the glassworks and residential areas in Sandomierz.

This subcomponent will include a number of works in the Raba sub-basin that are selected using an optimized investment variant (investment scenario) which prioritizes those investments that increase retention of water upstream and/or combine social and technical feasibility with the highest economic returns, that is, those that generate demonstrated benefits at local scale, in the downstream stretch, as well as further downstream along the Upper Vistula

The Phase 1 investments are those that have been confirmed on technical and economic merit, and whose functioning does not depend on the selection of the other structural elements and other measures (notably, the change in the operational schedule of the Dobczyce Reservoir) in the basin.

The subcomponent will finance the Phase 1 investments, and the full preparation of the Phase 2 investments, to bidding level.

This subcomponent will include a number of works in the San sub-basin that are selected using an optimized investment variant (investment scenario) which prioritizes those investments that increase retention of water upstream and/or combine social and technical feasibility with the highest economic returns, that is, those that generate demonstrated benefits at local scale, in the downstream stretch, as well as further downstream along the Upper Vistula.

The Phase 1 investments have been confirmed on technical and economic merit, and their functioning does not depend on the selection of the other structural elements and other measures (notably, th construction of flood retention reservoirs and dry polders) in the basin.

The Phase 1 investments concern the Sanozcek dike (1.61 km), and the construction of the Golcowa dry polder (Ropa River). The subcomponent will finance the Phase 1 investments, and the full preparation of the Phase 2 investments in the three sub-basins, to bidding level

The EC intensively monitors and assesses this development, is able to fund capacity building, and disposes of financial incentives to the reform.

Poland, EC and co-financiers have confirmed their preference of technical and operational leadership by the Bank for new large investment programs with the partial aim to use the programs to bring the institutional innovations into practice and pilot initiatives that are less well covered under EC support, such as the conversion of RBMPs into realistic investment programs.

Strengthening of the flood forecasting system and the operations centers will contribute substantially to the flood reducing benefits obtained with the construction of dry polders for flood management in Poland under the current and all previous investment schemes.

Principal investments include: (a) improvements/replacements at automatic weather stations; (b) replacing tipping bucket rain gauges by more reliable laser gauges; (c) installing snow depth sensors; (d) upgrading radio communication systems; (e) equipping some stations with water quality monitoring facilities; and (f) various installation and construction works.

Principal investments include: (a) IT platform extension; (b) development of the national flash flood forecasting system; and (c) catchment indexing and rainfall-runoff model development.

This work package comprises the following subtasks: (a) installation of monitoring and telemetry equipment at reservoirs, dry polders, and critical dam/dike sections; (b) modernization and implementation of local control systems for selected hydraulic structures; (c) improvement of equipment for guiding the inland navigation on the Upper, Free-flowing and Lower Odra; (d) modernization of field/mobile units; (e) installation of equipment at the operations and crisis centers to secure optimal provision of services; and (f) measures to assure reliable functioning and resilience of all essential installations).

The PCU will dispose of a modest budget for the equipping of offices and for incremental operating costs.

Procurement of works

The rehabilitation, reconstruction, and other large civil works will be procured through the International Competitive Bidding (ICB) procedure, following prequalification.

Procurement of goods and non-consulting services.

	IB
	Am
I. Investment Costs	
A. Civil Works	2
B. Land Acquisition and Resettlement Costs	
C. Design Costs	
D. Supervision of Construction	
E. Studies	
F. Goods	
G. Operating Costs	
Total PROJECT COSTS	4
Front-end fees	
Total Disbursement	4

As the basins have evolved into economic growth the increasing economic damages now warrant the costs of increased protection levels.

One of the reasons for this rise is the overload of flood-prone areas with settlements, industrial parks, transport corridors, etc.

The project would also further strengthen the national flood forecasting and operational capability (for existing and new infrastructure such as locks, weirs, barrages, reservoirs, etc

A fundamental part of the economic appraisal is the estimation of the expected benefits to be attained by the project investments.

Given that in Poland investment cost levels are comparable to those in Germany, but operational costs (primarily staffing expenditure) typically only half, a ratio of 0.5% is an acceptable estimate for Poland.

As the economy is expected to continue its growth, the additional fiscal burden to maintain and operate the new infrastructure from the time of completion of the works around 2020 or thereafter, can be, in principle, well accommodated in the growing budgets of the Implementing Agencies.

The Flood Risk Management Plans would include incentives to induce higher budgets for this purpose.

Damage to public infrastructure and facilities included losses to municipal infrastructure, public buildings (schools, health facilities, etc.

Induced economic benefits.

This assessment assumed the multiplier at 2.5 as recommended for developed economies by the American National Bureau of Economic Research after analyzing the effects of governmental investments in 44 countries (20 developed and 24 developing economies) in the years 1960-2[ta]. The induced benefits were estimated using this multiplier and the average net profit for the Polish business investments which was estimated at 4.6 percent in the period from January to June 2[is].

The proposed investment will also have an enabling environment for increasing the tourism income flowing into some of the proposed protected areas, as the risk of floods is reduced and the safety for tourists is increased.

These benefits were quantified taking into account the current number of tourists per area and the cost of stay per day for the average traveler (approximately PLN 9[be]).

Benefits of PLN 32.17 million per year were estimated by multiplying the expected increase of transported cargo by the reduction in transport costs per ton/km through the rive[ed].

Exploitation of aggregate mining. The Project is expected to generate also benefits from the exploitation of gravel and other materials for construction in the proposed reservoir areas.

Thanks to effective EWS, communities gain time to react moving to higher ground, out of the floodplain; elevating valuables to a higher floor or moving property such as cars outside of the floodplain; and/or building temporary walls with sandbags to keep water out of a structure or a property.

The assessment was carried out by comparing the incremental capital and operating costs require with the expected incremental benefits expected to result from the implementation of the project. The evaluation considered a 50-year period and two different alternatives for the estimation of the results: using a discount rate of (i) 5 percent (following the Guide to Cost-Benefit Analysis of Investment Project, European Commission, 2008, as well as OECD guidance); and (ii) 10 percent (as the minimum usually applied by the World Bank for this type of analysis).

The economic feasibility of the project and of each of its main components was measured through estimation of the economic rate of return (ERR) and net present values (NPVs) based on the above described assumptions As can be seen in Table 5.1, the estimated overall project ERR is 13.5 percent, which is well in excess of the long term opportunity cost of capital (OCC).

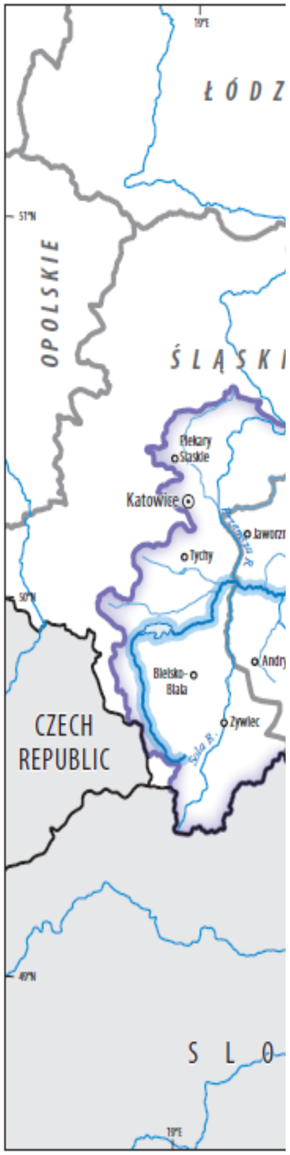
The experience with the ongoing Odra River Basin Flood Protection Project has shown that the government is committed to provide the budget for investment as well as for proper O&M.

The following assumptions were selected for testing variations and the sensitivity of the expected results: (i) investment costs; (ii) O&M costs; (iii) benefits from material and non-material avoided-flood damages; and (iv) the induced economic benefits.

The analysis shows that the estimated ERR is robust and not very sensitive to changes in costs and/or benefits: even a 30 percent increase in investment costs, or up to 40 percent reduction of expected benefits from avoided material flood damage would maintain the ERR above 10 percent.

Several policy review documents and technical papers including economic analysis of sector projects co-funded by the International Financial Institutions recommend an increase of the current levels of funds available for O&M of the flood protection structures. The issue is attracting growing interest of national policy makers as well as awareness about the need for higher O&M allocations for existing and new facilities given the increased frequency and intensity of flood events.

14°
Baltic Sea
54°
ZACHODNIOPOM
Szczecin
1.A
1.B
Gorzów
Wielkopolski
1.C
Słubice
LUBUSKIE
52°
Zielona
Góra
GERMANY
Neisse
DOLNOŚL
50°
CZECH REPU
14°
16°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Soła R.
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